

Jumarald Pharmacy & Wellness

Full Technical Architecture, Security & Operational Documentation — v3.0.0

1. Executive Summary

Jumarald Pharmacy & Wellness is a state-of-the-art digital healthcare platform operating in Ghana. The system unifies e-commerce pharmaceutical sales, certified prescription validation, telehealth physician appointments, real-time multi-branch inventory tracking, and an intelligent **Superintendent Clinical AI Assistant ('Dr. Jumarald AI')**.

The platform complies strictly with the Food and Drugs Authority (FDA Ghana), Pharmacy Council regulations, and the Ghana Data Protection Act 2012 (Act 843).

Superintendent Pharmacist: Pharm. Philip Bruce-Tagoe (GPHC Reg. No. 2050984).

2. System Architecture & Tech Stack

The solution is deployed as a modular, high-performance monorepo:

Component	Technology Stack	Key Features
Frontend Storefront	Next.js 15, React 19, Tailwind CSS	Patient portal, catalog, AI chat widget, reviews
Admin Console	Next.js 15, Recharts, Lucide Icons	Operational management, global inventory reports
Backend API	Express.js 4, TypeScript, Socket.io	REST API endpoints, WebSockets, Paystack webhooks
Database & ORM	Neon PostgreSQL, Prisma ORM	Relational data, schema safety, ACID transactions
Clinical AI Engine	OpenAI GPT-4o-mini / Gemini 1.5	Autonomous intent classification & multi-tool execution
Security & Auth	JWT, bcrypt, Zod, Rate Limiters	RBAC, token validation, resource rate limits

3. Superintendent AI Assistant ('Dr. Jumarald AI')

Dr. Jumarald AI provides clinical triage and product recommendations. It incorporates autonomous intent classification, structured tool execution, and a high-reliability rule-based fallback engine:

- **searchProducts:** Multi-keyword catalog query returning live prices in GH₵ and availability.
- **getProductDetails:** Retrieves dosages, indications, and active ingredients.
- **checkDrugInteraction:** Pharmacological contraindication analysis across multiple active medications.
- **findNearbyBranches:** Locates physical branches, operating hours, and pickup details.
- **getUserOrders:** Fetches patient order tracking status for authenticated sessions.
- **createPharmacistConsultation:** Escalates complex cases to the Superintendent Pharmacist queue.

4. Security Architecture (CIA Triad)

4.1 Confidentiality: Stateless JWT sessions, bcrypt password hashing, and active user verification in authenticateToken middleware.

4.2 Integrity: Prisma ACID transactions for inventory updates, Zod input validation, and Paystack HMAC SHA-512 signature verification.

4.3 Availability: Rate limiters applied to /api/v1/auth (20/15m), /api/v1/ai (20/m), /api/v1/prescriptions (30/15m), and /api/v1/orders (120/m).

5. Regulatory Compliance Suite

- **Acceptable Use Policy (/acceptable-use):** Rules against prescription forgery, identity theft, and prompt-injection.
- **Terms of Service (/terms):** Prescription verification guidelines, cold-chain transport, and non-returnable drug rules.
- **Privacy Policy (/privacy):** Patient rights and data protection standards under the **Ghana Data Protection Act 2012 (Act 843)**.